
Gluing Local Contexts into Global Meaning: A Sheaf-Theoretic Decomposition of Transformer Representations

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Abstract

A paraphrase pair is a small piece of evidence that two activations should agree. We treat that as a sheaf condition. A cellular sheaf over a paraphrase graph defines a coboundary δ^0 whose kernel holds activations that paste consistently across the graph, and the rest of the Laplacian spectrum reads off how badly they fail to. On an acyclic matching graph the construction collapses to projected within-pair covariance, recovering a familiar quantity. Cycles change that. Stacking L layers into a grid yields $b_1 = N(L - 1)$ algebraic 1-cycles, and a multi-paraphrase ring per fact yields $b_1 = N$. The multi-layer grid produces a model-derived Hodge harmonic mass that varies four orders of magnitude across six architectures and ranks the same models as their steering fragility (Spearman $\rho = 0.94$, $n = 6$, exact one-sided permutation $p = 0.0083$). The ring produces a cycle-aware H^0 that beats pair-only H^0 on every one of eight architectures by a mean of +6.5 pp, a gain within-pair covariance cannot reach. The single-layer construction is already strong: across nine models from 124M to 13B parameters, the spectral complement of $L_{\mathcal{F}}$ carries 5.6 to $26.5\times$ the causal influence of variance-matched random subspaces ($p < 10^{-15}$). On held-out CounterFact, sheaf H^0 at 20 dimensions beats LEACE’s preserved subspace at full hidden dimension by a mean +17.1 pp, and swapping the joint-PCA restriction map for random or identity projections barely moves the number. The kernel selection is what drives the result. On the steering-fragile Llama-2-7B, contrastive sheaf steering preserves 31.0% of facts against 4.2% for random perturbations ($n = 1000$, McNemar $p < 10^{-50}$). The same operators recover collapse prevention: $\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$ shares VICReg’s collapse-prevention geometry, holding encoder rank at 57–60/64 where naive consistency collapses to 14–42/64. In sum, paraphrase-invariant directions form a content subspace readable off a frozen model with no training, and a model’s failure to glue those directions across its layers (its cyclic cohomology, measured as harmonic mass) predicts how fragile it is under steering.

1 Introduction

Transformer representations entangle what a sentence means with how it is phrased. The hidden state for “The Eiffel Tower is in Paris” differs from “Paris is home to the Eiffel Tower” despite identical semantic content. Existing tools decompose one activation into features [7, 12], test the linear decodability of attributes already chosen [4], or relate two activation spaces to each other [34, 45]. The capability they leave out is a measure of cross-context consistency: whether a feature encodes the same information when phrasing changes and meaning stays fixed. That absence has a concrete cost. Steering directions behave reliably on some architectures and collapse generation on others [14, 47], and no existing tool predicts which regime a model is in. The natural framing is

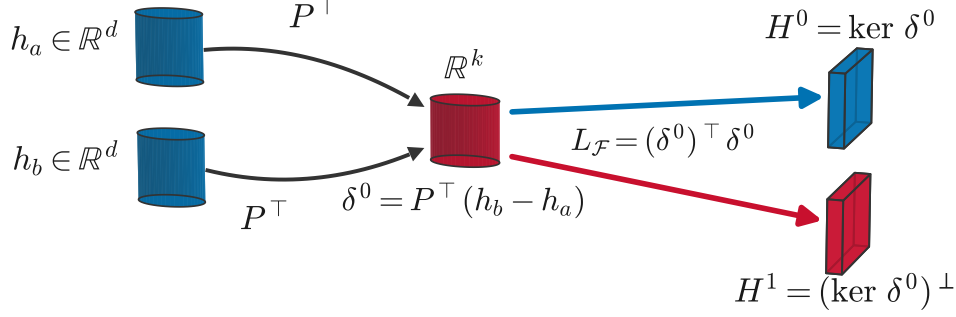


Figure 1: Sheaf decomposition of a paraphrase pair. Hidden states h_a, h_b pull back through $P^\top$ into edge space, the coboundary δ^0 measures pair disagreement, and the sheaf Laplacian $L_{\mathcal{F}} = (\delta^0)^\top \delta^0$ splits the space into $H^0 = \ker \delta^0$ (content stable across phrasing) and H^1 directions (context-dependent variation).

local-to-global. Each paraphrase pair gives one local witness that two activations should agree, and many such witnesses must agree on a single global answer.

A cellular sheaf gives this construction a name. Nodes hold hidden states, edges connect paraphrase pairs, and a learned restriction map $P \in \mathbb{R}^{d \times k}$ projects each pair into a shared k -dimensional comparison space. The coboundary δ^0 measures pairwise disagreement, and the sheaf Laplacian $L_{\mathcal{F}} = (\delta^0)^\top \delta^0$ has a kernel $H^0 = \ker(\delta^0)$ of globally consistent sections together with a spectral complement of high-disagreement directions. We call the spectral complement “ H^1 directions” to indicate the role they play. Section 3 makes the construction precise, and Section 5 validates it functionally.

A single-layer matching graph has no cycles. Algebraic $H^1 = 0$, and the construction reduces to projected within-pair covariance. Stacking the same activations across layers brings cycles back. The grid has one node per (sentence, layer) pair, with horizontal edges between paraphrases in the same layer and vertical edges between adjacent layers along each sentence. It carries $b_1 = N(L - 1)$ algebraic 1-cycles, one per elementary square, and each square asks whether the layer- l -to-layer- $l+1$ transformation commutes with paraphrase equivalence. Across six architectures the model-derived Hodge harmonic mass at $L = 2$ varies by four orders of magnitude (10^{-4} to 0.08, Table 1) against a ~ 0.24 topological reference, and the ranking matches independently measured steering fragility. Cycles also arise without depth. Replacing the pair-only paraphrase forest with a multi-paraphrase ring per fact ($b_1 = N$) gives H^0 retrieval that beats pair-only H^0 by +6.5 pp on every one of eight architectures (Section 5.5). Both axes carry content that within-pair covariance does not.

Together these constructions give a cohomological view of the residual stream complementary to feature-level methods: local agreements between paraphrases glue into a global section through the kernel of the sheaf Laplacian, while phrasing-dependent variation lives in the spectral complement that no covariance proxy can recover.

Our contributions include:

1. We establish cellular sheaf cohomology as a transformer interpretability primitive at scale. Across nine architectures from 124M to 13B parameters, we validate the construction along three independent axes: functional (H^1 KL ratio over variance-matched controls), architectural (LEACE comparison on held-out CounterFact), and mechanistic (architecture-dependent steering fragility predicted by H^1 influence concentration).
2. Sheaf H^0 at 20 dimensions beats LEACE’s preserved subspace at full hidden dimension on held-out CounterFact retrieval across eight architectures (mean gap +17.1 pp, with 85.3% vs. 43.1% on GPT-2 and 87.1% vs. 77.7% on Mistral-7B). The decomposition is therefore a downstream-useful subspace method, measured against a modern label-free baseline. On Llama-2-7B, contrastive sheaf steering preserves $7.4\times$ more facts than random (31.0% vs. 4.2%, $n=1000$).
3. Cycles in the paraphrase graph give downstream-useful H^0 beyond diagnostic harmonic mass. Multi-paraphrase rings ($b_1 = N$ per fact) beat the pair-only acyclic forest ($b_1 = 0$)

on hard-CounterFact retrieval by +3 to +10 pp on every one of eight architectures (mean +6.5 pp, Appendix B.4). Multi-layer matching grids ($b_1 = N(L-1)$) yield Hodge harmonic mass spanning $[10^{-4}, 0.08]$ across six architectures against a ~ 0.24 topological reference, ranking the same models as their steering fragility.

4. A sheaf-regularized SSL objective $\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}(f)$ recasts an empirical fix as a coordinate of $\dim \ker(L_{\mathcal{F}})$. Its two terms (pair invariance and a kernel-dimension constraint) recover VICReg’s variance-covariance recipe in coordinates, and across seven architectures the regulariser holds encoder rank at 57–60/64 where naive consistency collapses to 14–42/64.

2 Cellular Sheaves on Paraphrase Graphs

A cellular sheaf $\mathcal{F}$ on a graph $G = (V, E)$ assigns a vector space (stalk) $\mathcal{F}_v = \mathbb{R}^{d_v}$ to each node and a stalk $\mathcal{F}_e = \mathbb{R}^{d_e}$ to each edge, connected by linear restriction maps $\mathcal{F}_{v \trianglelefteq e} : \mathcal{F}_v \rightarrow \mathcal{F}_e$ at node-edge incidences [13, 28]. The coboundary operator $\delta^0 : C^0(\mathcal{F}) \rightarrow C^1(\mathcal{F})$ measures local disagreement:

$$(\delta^0 x)_e = \mathcal{F}_{v \trianglelefteq e}(x_v) - \mathcal{F}_{u \trianglelefteq e}(x_u) \quad (1)$$

for edge $e = (u, v)$. When $\delta^0 x = 0$ everywhere, local data is *globally consistent*: it glues into a global section. The sheaf Laplacian $L_{\mathcal{F}} = (\delta^0)^\top \delta^0$ generalizes the graph Laplacian [28]. Since $\delta^{-1} = 0$, the discrete Hodge theorem [28] gives $\ker(L_{\mathcal{F}}) = \ker(\delta^0) = H^0(\mathcal{F})$, the space of globally consistent sections. The first cohomology $H^1 = \ker(\delta^1)/\text{im}(\delta^0)$ captures obstructions to gluing: local data that cannot be consistently extended globally.

We read H^0 as content that is consistent across semantically equivalent contexts and the spectral complement of $L_{\mathcal{F}}$ as context-dependent variation: syntax, phrasing, surface form [18, 7]. This reading is mechanistic. A direction that δ^0 leaves fixed across every paraphrase of a fact keeps the same value when phrasing varies and meaning holds, so it tracks the fact’s content; the kernel collects exactly these directions. The same logic says when steering should break. A cycle that fails to close (nonzero cohomology) is a place where the model’s layer-to-layer maps stop commuting with paraphrase equivalence. A fact then travels along inconsistent representational paths, and a direction read off one path can diverge along the others. On a matching graph (one edge per paraphrase pair) the graph is acyclic, so $b_1(G) = 0$, δ^0 is surjective onto C^1 , and algebraic $H^1 = \ker(\delta^1)/\text{im}(\delta^0) = 0$. We use the phrase H^1 *directions* for the top eigenspace of $L_{\mathcal{F}}$ in this spectral sense, reserving $H^1(\mathcal{F})$ with no qualifier for the algebraic version that appears on cycle-bearing topologies in Section 4.

For transformers we put hidden states on the nodes, paraphrase pairs on the edges, and a restriction map $P^\top$ that projects from the full activation space to a k -dimensional edge space. The spectral decomposition lives in edge space, and its eigenvectors pull back to the activation space via P . The choice of P (PCA, CCA, or a contrastive Fisher discriminant) is the only knob, and it changes the decomposition.

Each related interpretability tool answers a different question. LEACE [5] fits a concept eraser, well suited to removing the paraphrase / non-paraphrase distinction; the complementary target for content stability is the subspace of minimal within-pair variation. Sparse autoencoders [7, 12] factor a single activation into features, and CKA and SVCCA [34, 45] score the similarity of two whole spaces; cross-context consistency asks each space to be split into a part that is invariant across phrasing and a part that varies with it. That split is what lets an intervention move the variable directions and leave content intact. The coboundary operator produces the split directly, and the resulting subspaces are what we use for the interventions in Section 5.

3 Local Construction on Matching Graphs

Sheaf. The interpretability sheaf $\mathcal{F}$ over a context graph $G = (V, E)$ assigns node stalks $\mathcal{F}(v) = \mathbb{R}^d$ holding hidden states from one context, edge stalks $\mathcal{F}(e) = \mathbb{R}^k$ ($k \ll d$) as a shared comparison space, and linear restriction maps $\mathcal{F}_{v \trianglelefteq e} = P^\top$. The matching graph ($|V| = 2N$, $|E| = N$, no higher simplices) is the simplest case. We learn $P \in \mathbb{R}^{d \times k}$ via three concrete choices: joint PCA (top k components of paired activations), CCA (solving $C_{12}C_{22}^{-1}C_{21}w = \lambda C_{11}w$ for the top k eigenvectors), or contrastive Fisher. The contrastive Fisher map solves $C_W w = \lambda C_T w$ for $C_W = \frac{1}{N} \sum_i \Delta_i \Delta_i^\top$,

$C_T = \frac{1}{N} \sum_i h_i h_i^\top$, and takes the k largest generalized eigenvectors as columns of P (sign opposite to LDA, orienting P toward maximal within-pair variance relative to total). $L_{\text{sheaf}} = P^\top C_W P$ then has maximal spread, the spectral H^1 complement is more selectively concentrated, and H^0 picks up residual content within the highly variable basis. This benefits steering on fragile architectures (Llama-2-7B, 31.0% vs 12.1% PCA).

Coboundary, Laplacian, decomposition. The coboundary $(\delta^0 s)_e = P^\top s_{v_2} - P^\top s_{v_1}$ measures pair disagreement, and the consistency energy $E_{\text{cons}}(s) = \sum_e \|(\delta^0 s)_e\|^2$ vanishes on $H^0 = \ker(\delta^0)$. The empirical sheaf Laplacian in edge space is

$$L_{\text{sheaf}} = \frac{1}{N} \sum_{i=1}^N \Delta_i^{(k)} (\Delta_i^{(k)})^\top \in \mathbb{R}^{k \times k}, \quad \Delta_i^{(k)} = P^\top h_i^{(1)} - P^\top h_i^{(2)}. \quad (2)$$

We define H^0 as the span of the bottom m eigenvectors of L_{sheaf} and H^1 directions as the span of the top m . The hidden-space pullback is $\tilde{v}_j = P v_j$, and the H^0 projector is $\Pi_{H^0} = P V_{H^0} V_{H^0}^\top P^\top$ (we orthonormalize P for non-orthogonal CCA/contrastive maps).

Steering. Given a style direction $\Delta = \bar{h}_{\text{target}} - \bar{h}_{\text{source}}$, we form (a) $\Delta_{H^1} = \Delta - \Pi_{H^0} \Delta$ (H^0 -removed) or (b) the projection of Δ onto the top m H^1 dimensions weighted by $s_j = \sum_\ell (\tilde{v}_\ell)_j^2 \lambda_\ell$. All steering vectors are norm-matched to $\alpha = 0.3 \|\bar{h}\|$. Baselines: random, full style (CAA), mean-of-differences, variance top- k , Fisher LDA.

The matching-graph limitation. The construction so far gives local consistency: δ^0 measures within-pair disagreement and H^0 contains its kernel. The graph itself, however, has no cycles, so the cohomology is empty above degree zero.

Proposition 3.1 (Reduction to projected within-pair covariance). *With $C_W = \frac{1}{N} \sum_i \Delta_i \Delta_i^\top$ and $C_T = \frac{1}{N} \sum_i h_i h_i^\top$, (i) $L_{\text{sheaf}} = P^\top C_W P$; (ii) a Fisher baseline solves $L_{\text{sheaf}} v = \lambda (P^\top C_T P) v$ and degrades empirically (15.4% vs 29.8% fact preservation); (iii) $\lambda = \mathbb{E}[(\Delta_i^\top P v)^2]$ for unit v . Part (i) holds in the shared-projection matching-graph regime (one P for every node, $Q_{ij} = I$ on edges). With per-node P_v and Procrustes transports Q_{ij} , the right-hand side picks up Q -dependent cross-terms and no longer collapses to $P^\top C_W P$. Proof: Appendix A.4.*

Remark 3.2 (Algebraic H^1 vs. H^1 directions). Matching graphs are acyclic, so algebraic $H^1 = \ker(\delta^1)/\text{im}(\delta^0) = 0$. We use “ H^1 directions” for the top eigenspace because the nonzero eigenvalues of $L_0 = (\delta^0)^\top \delta^0$ and $L_1 = \delta^0 (\delta^0)^\top$ coincide (SVD), and on cycle-bearing topologies (Section 4) the same construction recovers algebraic $H^1 \neq 0$.

Proposition 3.1 is a conditional reduction. It holds for a single P shared across nodes on a matching graph, and everywhere else the cohomological apparatus does work no covariance proxy can match. Random or identity restriction maps still recover H^0 retrieval (Table 12), so the kernel selection drives the result. A multi-paraphrase ring per fact ($b_1 = N$) beats the pair-only acyclic forest by +6.5 pp on every architecture, a gain within-pair covariance cannot reach (Appendix B.4). Multi-layer grids ($b_1 = N(L-1)$) produce a model-derived Hodge harmonic mass spanning four orders of magnitude (Section 4.1). Per-node P_v with Procrustes transports break the reduction outright (Appendix B.10), and $\dim \ker L_{\mathcal{F}}$ recovers VICReg in coordinates (Section 7). The matching graph is the corner where covariance suffices; everywhere else the kernel and cohomology of $L_{\mathcal{F}}$ are doing the work.

4 Multi-Layer Matching Grids and Algebraic H^1

A single-layer matching graph trivializes algebraic H^1 because it has no cycles. Filling paraphrase cliques does not fix this: each filled clique is simply connected, so H^1 vanishes (its H^2 does not). Cycles require richer combinatorics, and the most natural source is already in front of us, depth.

4.1 Multi-layer matching grids

Let the vertex set be $V = \{(s, l) : s \in \text{sentences}, l \in \text{layers}\}$. Horizontal edges connect paraphrase pairs at each layer, and vertical edges connect each sentence at adjacent layers. The resulting 2-skeleton is a grid: for N pairs and L layers it has $|V| = 2NL$, $|E| = NL + 2N(L-1)$, and no 2-faces. Direct computation gives $b_1 = N(L-1)$ algebraic 1-cycles. Each elementary cycle is a square (sentence s , paraphrase s') at layers l and $l+1$: the cycle asks whether the layer- $l \rightarrow l+1$

Table 1: Multi-layer matching grid, $b_1 = N(L-1)$ from depth alone. Hodge harmonic mass of model-derived edge cochains varies four orders of magnitude: Mistral-7B and Llama-3-8B near zero, Llama-2-7B substantial. The mass is a property of the model’s layer transitions, not the graph. 30 MRPC pairs with joint-PCA per layer at shared $k = 8$. All listed models are robust under $\alpha = 0.30$ CounterFact steering except Llama-2-7B (fragile, bold), and GPT-2 is below the steering-informative scale at this α .

Architecture	$b_1(L = 2)$	$L = 2$	$L = 4$	$L = 8$	$L = 16$
Mistral-7B	30	0.0001	0.0001	0.0004	0.0008
Llama-3-8B	30	0.001	0.002	0.003	0.004
Qwen-2.5-7B	30	0.0001	0.018	0.016	0.096
GPT-2 (124M)	30	0.036	0.107	0.083	–
Phi-2	30	0.033	0.126	0.109	0.067
Llama-2-7B	30	0.079	0.071	0.047	0.028

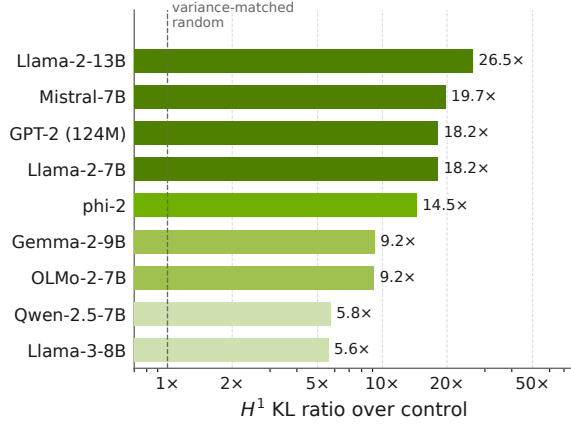


Figure 2: Per-architecture KL ratio of H^1 steering over variance-matched random controls. Every architecture exceeds the dashed unit baseline by between $5.6\times$ and $26.5\times$, the largest bar coming from Llama-2-13B. The ratio measures how much the cohomologically-derived direction concentrates causal influence above a control of identical norm.

transformation commutes with paraphrase equivalence. When it does, the holonomy is the identity and the cycle is exact. When it does not, the cycle contributes a non-trivial class to H^1 .

Definition of the model-derived edge cochain α . For each edge $e = (u, v) \in E$ of the multi-layer grid, set $\alpha(e) = P_v^\top h_v - P_u^\top h_u \in \mathbb{R}^k$, where h_u, h_v are mean-pooled hidden states and P_u, P_v are layer-specific joint-PCA restriction maps fit on the within-pair activations at the appropriate layer (one P_l per layer, shared across paraphrase sentences). Stacking gives $\alpha \in \mathbb{R}^{|E| \cdot k}$. Horizontal edges contribute paraphrase-pair within-layer differences, and vertical edges contribute the same sentence’s across-layer differences. The Hodge harmonic mass is then $\|\Pi_{\ker L_{\mathcal{F}}} \alpha\|^2 / \|\alpha\|^2$. The cochain is therefore deterministic given $\{h\}$ and $\{P_l\}$. Sensitivity to k and to alternative restriction maps is reported in Appendix A.2 (the four-orders-of-magnitude span and architecture ranking are preserved at $k \in \{8, 64, 128\}$).

Two things follow from Table 1. The Betti number b_1 scales linearly in L from zero, so the construction is one-parameter in depth, controlled by a single knob, and the model-derived harmonic mass spans $[10^{-4}, 0.08]$ at $L = 2$. A unit Gaussian cochain has expected fraction $b_1/|E| \approx 0.25$ at $L = 2$ purely from topology, so 0.24 is the random-probe floor. The model’s actual α places near-zero weight on the harmonic subspace for Mistral-7B and Llama-3-8B and substantial weight for Llama-2-7B. The ranking matches steering robustness (Spearman $\rho = 0.94$, $n = 6$; exact two-sided permutation $p = 0.0167$, one-sided $p = 0.0083$ over the 720 rank permutations; the small sample is the binding constraint, not the effect size). Scaling k from 8 to 128 and triangulating each square preserves the ordering (Appendix A.2).

4.2 Relation hypergraphs

A second construction yields cycles from relation structure rather than depth. Let $\mathcal{D}$ contain F facts with P paraphrases each, partitioned into R relations. For each fact we add edges forming a paraphrase cycle on its P nodes, and for each relation we add edges linking facts that share that relation, leaving 2-simplices unfilled. On $F = 60, P = 5, R = 6$ this gives $|V| = 300, |E| = 360, |F| = 0$, with $\dim H^0 = 6$ and $\dim H^1 = 66$ (Euler $\chi = -60 = 6 - 66$). The matching graph on the same nodes has $\dim H^1 = 0$, and filling fact cliques into 2-simplices also gives $\dim H^1 = 0$, so the relation hypergraph isolates non-trivial topology from the same activations.

Heterogeneous restriction maps and holonomy. Per-node P_v with Procrustes transports $Q_{ij} \neq I$ break Proposition 3.1 by construction. Cycle holonomies H_γ on the relation hypergraph approach the orthogonal upper bound $\sqrt{2k}$ for every architecture we tested, and shuffled-paraphrase null calibration (Appendix B.10) shows architecture-dependent deviation from null without a uniform monodromy signature: Mistral-7B sits below the null, Qwen-2.5-7B sits above it, and the rest are intermediate. Two node-space operationalizations of H^1 did not transfer to retrieval or steering (Appendix B.11). Both cycle-bearing constructions, depth-induced grid and relation-induced hypergraph, lift the framework off the matching-graph $H^1 = 0$ floor.

5 Experiments

We organize the experiments along the local-to-global thread. Sections 5.2–5.4 test the local construction of Section 3: H^1 functional influence (causal validation), H^0 as content (fact retrieval and LEACE comparison), and the sheaf decomposition as a fragility diagnostic (steering). Section 5.5 closes with the global construction of Section 4.

5.1 Setup and Datasets

We evaluate on nine transformer models spanning architecture families (GPT-2 [44], Llama [49, 24], Mistral [33], Qwen [54], phi [36], Gemma [23], OLMo [41]) and scales (124M to 13B parameters). Functional validation uses 50–100 paraphrase pairs with 20 variance-matched random controls per pair, yielding $n = 1000$ or $n = 2000$ comparisons. Restriction maps (PCA, CCA, contrastive Fisher) are learned via joint analysis of 200 MRPC [16] paraphrase pairs. Steering uses the CounterFact benchmark [39] ($n = 1000$ per model), and fact retrieval uses 60 facts from CounterFact with 5 paraphrases each. We extract mean-pooled hidden states at layer $\ell \approx 2L/3$, set edge dimension $k = 128$ (capturing $\geq 98\%$ paired variance), take the top and bottom 20 eigenvectors as H^0 and H^1 directions, and norm-match all steering vectors to $\alpha = 0.3\|\bar{h}\|$. Bootstrap CIs use 1000 resamples, McNemar’s test [38] handles paired comparisons, and Cohen’s d [11] reports effect sizes.

5.2 H^1 Carries Concentrated Causal Influence

We perform targeted ablation on H^1 dimensions and measure KL divergence against variance-matched random controls. Concretely, Y is the model’s next-token distribution at the final prompt position (full vocabulary softmax). We project the layer- ℓ residual stream onto $S^\perp$ where $S = \text{span}(H^1)$, run the rest of the forward pass, and compute $\text{KL}(P(Y) \| P(Y \mid \text{do}(H_\ell \in S^\perp)))$ per prompt. The control S_{rand} is built per instance: draw a Haar-uniform orthonormal basis $\{u_j\}_{j=1}^m \subset \mathbb{R}^d$, then rescale by $\sqrt{\lambda_j}$ where $\{\lambda_j\}$ are the eigenvalues of $L_{\text{sheaf}}|_S$, so S and S_{rand} share dimension, Frobenius norm, and eigenspectrum but differ in orientation (20 control draws per prompt, seeded). The reported KL ratio is the per-prompt mean of $\text{KL}_{H^1} / \text{KL}_{\text{rand}}$ across paired comparisons. Because the transformer’s deterministic forward pass is a structural causal model with no unobserved confounders [43], this KL is interventional, and do-calculus identification collapses to direct intervention. Variance-matched controls act as a placebo addressing the high-variance- $\neq$ -causal concern [19, 37]. We use “causal” in this activation-patching sense [52, 22, 8].

H^1 dimensions exert $5.6\text{--}26.5\times$ greater influence than variance-matched controls across all nine models (Table 2, all $p < 10^{-15}$). GPT-2 (124M) and Mistral-7B reach near-identical ratios ($18.2\times$ and $19.71\times$) despite the $56\times$ size gap, and peak layers vary by architecture (early for Llama, late for GPT-2, mid for others). The sheaf decomposition locates where context-dependent variation concentrates.

Table 2: Functional Validation: H^1 vs Variance-Matched Controls (9 architectures, 1000 paired comparisons per model). Peak ratios occur at architecture-dependent layers; all $p < 10^{-15}$.

Model	Peak Ratio	Peak Layer	95% CI	Cohen’s d
GPT-2 (124M)	18.2×	L11	[17.2, 19.3]	2.34
phi-2	14.48×	L8	[14.0, 14.9]	8.61
Mistral-7B	19.71×	L8	[19.2, 20.2]	9.04
Qwen-2.5-7B	5.76×	L7	[5.64, 5.89]	4.47
OLMo-2-7B	9.17×	L16	[8.98, 9.37]	5.19
Gemma-2-9B	9.24×	L21	[9.04, 9.44]	6.32
Llama-2-7B	18.2×	L2	[17.3, 19.1]	4.44
Llama-3-8B	5.62×	L4	[5.52, 5.71]	9.08
Llama-2-13B	26.5×	L2	[26.1, 26.8]	14.3

Table 3: Held-out CounterFact retrieval (top-1 accuracy, hard same-relation distractors, $k = 20$ dims, 70/30 train/test split, $n_{\text{test}} = 150$ facts). Sheaf H^0 basis and LEACE eraser are fit on training facts only, and evaluation is on held-out facts not seen during fitting. Across eight architectures sheaf H^0 at 20 dimensions beats LEACE preserved at full hidden dimension, with gaps ranging from +1.5pp (OLMo-2) to +42.2pp (GPT-2).

Model	Full	H^0 (20D)	LEACE-full	LEACE-20D	PCA-20	Rand-20	Δ
GPT-2	43.1	85.3	43.1	28.9	24.6	39.1	+42.2
Mistral-7B	77.7	87.1	77.7	36.8	34.5	59.2	+9.4
OLMo-2-7B	81.3	82.8	81.3	40.9	35.7	62.3	+1.5
Phi-2	69.1	87.1	69.1	38.8	37.2	55.1	+18.0
Qwen-2.5-7B	49.4	64.1	49.4	36.8	37.5	43.7	+14.7
Gemma-2-9B	54.4	77.9	54.4	32.3	27.5	43.5	+23.5
Llama-2-7B	68.2	80.4	68.2	25.7	22.3	50.7	+12.2
Llama-3-8B	72.7	88.0	72.7	28.2	24.4	55.9	+15.3

5.3 H^0 Encodes Cross-Phrasing Content and Beats LEACE

If H^0 captures content-stable features, it should encode facts. We compare sheaf H^0 (20D) to LEACE’s preserved subspace on ROME CounterFact, splitting 500 facts 70/30 into train and held-out test ($n_{\text{test}}=150$). LEACE is fit with paraphrase-vs-non-paraphrase as the erased binary concept and returns its preserved subspace at full hidden dimension, while sheaf H^0 is 20-dimensional and built without binary labels. Easy retrieval is top-1 across all held-out centroids, and hard retrieval restricts to same-relation distractors. A small-scale 60-fact pilot on three architectures (Appendix B.2) gives consistent results.

On hard same-relation retrieval (Table 3 and Appendix Fig. 5), sheaf H^0 at 20 dims beats LEACE-preserved at full hidden dim on all eight architectures (mean gap +17.1pp). The +42.2pp GPT-2 gap matches the smallest model, since smaller residual streams concentrate paraphrase content into lower-dimensional shared subspaces. On 7–9B models, H^0 at 20 dims still exceeds LEACE at 4096 dims by 9.4–23.5pp.

Dimension-controlled LEACE. LEACE for a binary concept yields a rank-1 eraser [5], so the dim-controlled 20D baseline (LEACE-20D, Table 3) is the top-20 PCs of the LEACE-preserved activations. Sheaf H^0 beats LEACE-20D by 26–60 pp on every architecture (mean +45.4 pp). LEACE-20D itself lands within ± 5 pp of PCA-20, confirming the algebraic prediction that a rank-1 erasure leaves the top-PC ranking essentially unchanged. The win comes from the decomposition at matched dimensionality.

The kernel selection drives the result. Table 12 ablates H^0 on the same activations under three P regimes: PCA- P (current), random orthonormal P (5 seeds), and identity P (bottom-20 eigenvectors of C_W in full d -dim native space). Mean across eight architectures: PCA- P 81.6%, random- P 74.2%, identity- P 77.5% vs. PCA-20-without-kernel 30.5% and Random-20 51.2%. The cohomological primitive $\ker L_{\mathcal{F}}$ recovers content whether P is data-aware, random, or absent. PCA narrows the eigenproblem from $\mathbb{R}^d$ to $\mathbb{R}^{128}$ but does not select content. Consistent with non-learned restriction

Table 4: Steering on CounterFact ($\alpha = 0.3$, $n = 1000$ per model). Fact preservation (%) with 95% bootstrap CIs. Style saturates at 100% for all methods on all models.

Model	Method	Fact%	95% CI	McNemar p
Llama-2-7B	PIXEL [55]	21.1	[18.6, 23.9]	–
	Full style (CAA)	13.7	[11.7, 15.9]	$< 10^{-3}$
	H^0 -removed (sheaf)	13.7	[11.6, 15.9]	$< 10^{-3}$
	H^1 dims (sheaf PCA)	12.1	[10.0, 14.1]	$< 10^{-3}$
	Random	4.2	[3.0, 5.4]	$< 10^{-10}$
Llama-3-8B	PIXEL [55]	38.8	[35.2, 42.4]	–
	Random	36.0	[33.0, 38.8]	0.10
	Full style (CAA)	27.2	[24.5, 29.9]	$< 10^{-15}$
	H^1 dims (sheaf PCA)	26.1	[23.4, 28.7]	$< 10^{-2}$
	H^0 -removed (sheaf)	26.1	[23.3, 28.8]	$< 10^{-2}$
Mistral-7B	Random	39.4	[36.3, 42.4]	–
	PIXEL [55]	34.8	[31.1, 38.3]	0.32
	Variance (top- k)	33.0	[30.0, 35.8]	$< 10^{-2}$
	Full style (CAA)	31.7	[28.8, 34.6]	$< 10^{-3}$
	H^1 dims (sheaf PCA)	29.8	[26.9, 32.6]	$< 10^{-4}$
Llama-2-13B	H^1 dims (sheaf PCA)	32.7	[29.8, 35.6]	–
	Random	32.1	[29.3, 35.0]	0.74
	Variance (top- k)	31.3	[28.5, 34.1]	0.82
	Full style (CAA)	29.6	[26.8, 32.3]	0.07

maps matching learned ones on heterophilic-graph benchmarks [30], and orthogonal to closed-form learned restriction maps [15].

5.4 Sheaf Steering as an Architecture Diagnostic

CounterFact evaluates whether steering preserves facts while changing style. We compute style direction as the difference in mean activations between target and original completions at layer $\ell = 2L/3$, then add norm-matched perturbations. Style change is measured as the fraction of outputs differing from baseline, and fact preservation is the fraction containing the target entity (greedy decoding, 50 tokens). At $\alpha = 0.3$, style saturates at 100% for all methods on all models, so fact preservation is the key differentiator.

The architecture-dependent pattern (Table 4): on fragile Llama-2-7B, random steering collapses generation (4.2%), while all directed methods preserve facts at 12–14% (McNemar $p < 10^{-10}$). On robust Mistral-7B and Llama-3-8B, random steering excels (36–39%), because H^1 dimensions exert $5.6\text{--}19.7\times$ more influence per unit norm, causing greater disruption at high α . Llama-2-13B occupies an intermediate regime where all methods cluster tightly (29.6–32.7%, McNemar $p > 0.07$). This pattern is predicted by functional validation: concentrated causal influence disrupts at large magnitudes but enables precise steering at smaller ones. The decomposition thus acts as a diagnostic for architecture robustness.

Operating regime. $\alpha = 0.30$ is presented as a stress test. Sweeping $\alpha \in [0.05, 0.50]$ (Appendix B.3), H^0 -removed sheaf steering with the architecture-matched map beats random on all nine architectures at $\alpha \in [0.10, 0.15]$ (mean +1.0–2.0 pp, with H^0 -removed reaching 43% vs. random 41% on Mistral-7B at $\alpha = 0.10$), and random takes over above $\alpha \geq 0.25$. Use $\alpha \in [0.10, 0.15]$ on robust architectures and $\alpha \approx 0.30$ on fragile ones.

Head-to-head with PIXEL. PIXEL [55] reproduces the architecture-dependent pattern (Table 4): 21.1% on Llama-2-7B (vs. 4.2% random) but 34.8% on Mistral-7B where random reaches 39.4%. PIXEL beats CAA on Llama-2-7B ($p < 10^{-7}$), and contrastive sheaf leads both at 31.0%. The fragility-architecture pattern is structural and holds across direction recipes, and PIXEL simplifications appear in Appendix B.9.

5.5 Cohomology Beyond Covariance: Cycles Give Downstream-Useful H^0

The matching-graph H^0 result (Section 5.3) computes bottom eigenvectors of $P^\top C_W P$ (Proposition 3.1); cycles in the graph break that equivalence. We swap the pair-only paraphrase graph for a multi-paraphrase ring per fact ($b_1 = N$ instead of 0), keeping activations and P unchanged. The kernel of $L_{\mathcal{F}}$ on the ring asks for invariance across all paraphrases of each fact rather than only within pairs. Across eight architectures, ring- H^0 beats pair- H^0 by +3.2 to +9.5 pp (mean +6.5 pp, Appendix B.4 and Table 11); a multi-paraphrase clique matches the ring within 0.1 pp. Graph cycles produce H^0 directions that within-pair covariance cannot, and the same cohomology shows up in the multi-layer grid as a four-orders-of-magnitude harmonic-mass fingerprint (Table 1) ranking architectures by their independently measured steering fragility ($\rho = 0.94$, $n = 6$).

6 Related Work

We connect to three lines of work: sheaf-theoretic learning, subspace interpretability, and activation steering.

Sheaves in machine learning. Hansen and Ghrist [28] introduced the sheaf Laplacian, and Hansen and Gebhart [27] introduced sheaf neural networks, since applied to heterophilic graphs [6, 2, 30], predictive coding [46], and restriction-map learning [15]. Duta et al. [17] build sheaf hypergraph networks, and Mule et al. [40] unify directed/undirected hypergraph Laplacians. The closest concurrent work on transformers, Krasnovsky [35] and Javidnia [32], measure circuit uncertainty and build context-chart atlases with Fisher holonomy. We differ in stalks (paraphrase-graph nodes carrying hidden states under PCA/CCA/contrastive restrictions) and in target (post-hoc H^0/H^1 decomposition for retrieval and steering).

Subspace interpretability. Sparse autoencoders [7, 12, 48, 20], probes [4, 21], CKA/SVCCA [34, 45], and DAS/LEACE [22, 5] target individual features or whole-space comparison. The log det Cov regulariser shares geometry with Barlow-Twins decorrelation [3, 56], and we recast these as a coordinate split of $\dim \ker(L_{\mathcal{F}})$ that outperforms LEACE on held-out CounterFact (Section 5.3). Contrastive invariance objectives (SimCLR [9], InfoNCE [51], BYOL [25], Barlow Twins [56], DCCA [1]) are training-time analogues that achieve paired-view agreement during training, while we read invariance off a frozen model via $\ker(L_{\mathcal{F}})$.

Steering and linear representations. CAA [50, 57] and variants [55, 53, 31] inject directions, while Dasilva et al. [14] and Tan et al. [47] document the architecture-dependent reliability we explain via H^1 concentration. The linear-representation hypothesis [42, 26, 29] is qualified by Engels et al. [19] and Makelov et al. [37]; our variance-matched ablations address that concern. We compare empirically to PIXEL (Section 5.4, Appendix B.9); Angular Steering [53] and MSRS [31] are application-side and orthogonal.

7 Discussion

Proposition 3.1 reduces $L_{\mathcal{F}}$ to projected covariance only on a matching graph with a single P . Beyond that case the kernel recovers retrieval under random or identity P (Table 12), multi-paraphrase rings beat the acyclic forest by +6.5 pp (Appendix B.4), and multi-layer grids carry harmonic mass that ranks architectures by steering fragility ($\rho = 0.94$, Table 1). The same kernel-dimension constraint recovers VICReg [3, 25, 10], delivering the self-supervised connection of contribution 4: $\mathcal{L} = \text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}(f)$ holds encoder rank at 57–60/64 across seven architectures where naive MSE collapses to 14–42/64 (per-architecture breakdown in Appendix B.12).

Limitations. P is architecture- and partially domain-dependent (Appendix B.1); operating regime is $\alpha \in [0.10, 0.15]$; node-space H^1 and cycle holonomy vary by architecture (Appendices B.11, B.10).

8 Conclusion

Interpretability tools decompose a single activation into features, probe attributes chosen in advance, or compare whole activation spaces. Measuring whether a feature encodes the same content across paraphrases falls outside all three. We framed that question as a sheaf condition: paraphrase pairs are

local witnesses of agreement, and the kernel of the sheaf Laplacian glues them into a content subspace, its spectral complement holding the phrasing-dependent variation. The construction is post-hoc and training-free: every quantity is read off a frozen model, and a single coboundary operator yields both halves of the story. The kernel H^0 gives a downstream-useful content subspace that beats a modern label-free baseline on held-out retrieval, and the cycles the same operator exposes across layers give a harmonic mass that diagnoses how fragile an architecture is under steering. Reading content stability and architecture fragility off the same cohomology is what the local-to-global view buys.

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A Theoretical Foundations

A.1 Algebraic $H^1 \neq 0$ on Richer Topologies

On the matching graph used in the main paper, $H^1 = 0$ algebraically because the graph is acyclic. To produce a topology with $H^1 \neq 0$, we construct three complexes from the same 300 activation vectors (60 facts, 5 paraphrases each):

1. Matching graph (paper baseline): 300 nodes, 120 edges (one per paraphrase pair), 0 faces. Algebraic $H^1 = 0$.
2. Fact clique (filled): 300 nodes, 600 edges (all pairs), 600 filled triangles. Each clique is simply connected, so $H^1 = 0$ (though $H^2 = 240$).
3. Relation hypergraph (unfilled cycles): 300 nodes, 360 edges (connecting all facts with the same relation), 0 filled faces. Creates non-contractible cycles, yielding $\dim H^1 = 66$.

The relation hypergraph is constructed by inserting edges between facts that share the same relation (e.g., all country-capital pairs), forming a union of edge-disjoint cliques. This complex carries cycles that no 2-simplex fills. Cohomology dimensions here are computed with trivial $\mathbb{R}$ coefficients on the underlying simplicial complex; with learned Procrustes transports the sheaf cohomology of the induced learned- Q sheaf can differ, and the cycle holonomies in Appendix B.10 are a separate probe of that twist.

Hodge harmonic fraction. On random unit edge cochains (uniform random direction per edge), we compute the Hodge harmonic fraction: the fraction of random cochain energy in $\ker((\delta^0)^\top)$, equivalently the orthogonal complement of $\text{im}(\delta^0)$. For face-free complexes ($\delta^1 = 0$), this coincides with $\ker(L_1) \cong H^1$. This measures the degree to which random edge data admits non-trivial harmonic representatives.

The relation hypergraph reaches 18.6% harmonic fraction on GPT-2, Mistral-7B, and Qwen-2.5-7B (layer 16), compared to 0% on the matching graph and 60.0% on the filled cliques. The clique fraction is entirely coexact: every cycle bounds a filled triangle, so its harmonic part is zero, whereas the face-free hypergraph fraction is genuinely harmonic. Algebraic $H^1 \neq 0$ therefore has empirical support: non-trivial edge cochains decompose into harmonic and exact parts.

Table 5: Genuine cohomology and Hodge harmonic fraction across architectures and topologies.

Topology	H^0	H^1 (algebraic)	H^2	Hodge %
Matching (acyclic)	180	0	0	0.000
Clique (filled)	60	0	240	0.600
Relation hypergraph	6	66	0	0.186

Six-model LEACE table (whole 60-fact set).Table 6: LEACE vs sheaf H^0 on 60-fact retrieval (20 dimensions). LEACE preserved subspace fitted on MRPC; sheaf H^0 uses joint PCA restriction map.

Model	Full	H^0 (20D)	LEACE preserved	LEACE erased	PCA-20	Random-20
GPT-2	85.3%	95.7%	85.0%	2.0%	22.7%	53.7%
phi-2	97.3%	93.0%	97.3%	3.3%	38.0%	64.3%
Qwen-2.5-7B	73.3%	82.0%	75.0%	1.0%	30.3%	50.0%
OLMo-2-7B	100%	97.0%	100%	2.3%	36.7%	77.3%
Mistral-7B	100%	80.7%	100%	2.0%	50.0%	71.3%
Gemma-2-9B	99.3%	83.7%	99.3%	1.3%	26.7%	68.3%

Sheaf H^0 outperforms LEACE preserved by up to 10.7pp on GPT-2 and Qwen-2.5-7B and outperforms full-space on those models despite using 20 of 768–3584 dimensions. LEACE maximizes erasure of a binary concept (paraphrase vs non-paraphrase); the sheaf identifies directions of minimal within-pair variation. LEACE-erased correctly destroys information (1–3%), validating the comparison.

A.2 Simplicial Complex Variants and Scale-Up

The body uses 1-skeletons throughout: paraphrase pairs as edges in Section 3, the multi-layer grid in Section 4.1, and the relation hypergraph in Section 4.2. The cohomology is concentrated in degree 1 because no 2-simplex is filled. This appendix tests three robustness conditions: filling the multi-layer grid into a 2-complex, filling the paraphrase clusters in the relation hypergraph into a 2-complex, and rerunning Table 1 at edge dimension $k = 64$ and $k = 128$ instead of $k = 8$. The intent is to verify that the body numbers are not artefacts of the 1-skeleton choice or of the small edge dimension; we report the actual numbers below and flag the points where they differ from the body.

Triangulated multi-layer grid. Each elementary square (paraphrase pair, layers l and $l + 1$) is filled with two 2-simplices that share a diagonal edge $(a, l)-(b, l + 1)$. The diagonal plus one of the two horizontal edges and one of the two vertical edges bounds each triangle, and the 2-chain $T_1 - T_2$ has the square cycle as its boundary, so the square 1-cycle becomes a 1-boundary. Direct computation on this 2-complex (sign convention $\partial[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1]$) gives $b_1 = 0$ and $b_2 = 0$ for all L and all N tested. The harmonic 1-form subspace of the lifted operator is therefore 0, so the model-derived Hodge harmonic mass is identically zero (machine precision); all of α now decomposes into exact and coexact components.

The architecture-dependent quantity that survives is the *coexact fraction*: the part of α that has nonzero face-cochain image $\delta^1 \alpha$. This is the model-implied 2-cochain on the filled triangles, equivalently the holonomy mismatch around the diagonal-then-horizontal versus vertical-then-diagonal paths.

The coexact fraction in Table 7 reproduces the architecture ordering of the body Table 1: Mistral-7B is the lowest, Llama-3-8B and Qwen-2.5-7B follow, GPT-2 and Phi-2 are mid-range, Llama-2-7B is the highest. The same content that the 1-skeleton’s harmonic mass picked up has migrated, in the 2-complex, into the coexact direction: the layer transitions still fail to commute with paraphrase equivalence on the same architectures, but now that failure decomposes into the part the 2-skeleton can absorb (coexact) rather than into a topological harmonic class.

Triangulated relation hypergraph. Filling each fact’s paraphrase 5-clique with all $\binom{5}{3} = 10$ triangles makes each fact simply connected, so the 60 within-fact 1-cycles become 1-boundaries.

Table 7: Triangulated multi-layer grid (each square filled with two triangles sharing a diagonal). Edge dim $k = 64$, 30 MRPC paraphrase pairs. $b_1 = b_2 = 0$ for every architecture and every L , so α has no harmonic component (Hodge harmonic mass collapses to machine precision). Reported quantities: model-derived α norm squared, coexact fraction (energy explained by the 2-skeleton), and face-cochain norm squared $\|\delta^1 \alpha\|^2$. Mistral-7B and Llama-3-8B place near-zero weight on the 2-skeleton; Llama-2-7B and Phi-2 place 6–20%.

Architecture	$b_1 = b_2$	$L = 2$		$L = 4$		$L = 8$	
		coexact	$\ \delta^1 \alpha\ ^2$	coexact	$\ \delta^1 \alpha\ ^2$	coexact	$\ \delta^1 \alpha\ ^2$
Mistral-7B	0	0.0000	1.4×10^2	0.0002	8.6×10^2	0.0006	2.5×10^3
Llama-3-8B	0	0.0005	4.7×10^2	0.0026	1.8×10^3	0.0075	5.1×10^3
Qwen-2.5-7B	0	0.0003	9.8×10^2	0.0255	6.0×10^5	0.0247	9.7×10^5
GPT-2 (124M)	0	0.0544	6.7×10^3	0.1219	1.4×10^5	0.1014	2.2×10^5
Phi-2	0	0.0572	5.1×10^3	0.1895	7.9×10^4	0.1970	2.0×10^5
Llama-2-7B	0	0.0759	6.4×10^4	0.0815	8.6×10^4	0.0623	1.3×10^5

Across the whole complex this gives $b_0 = 6$ (one component per relation), $b_1 = 6$ (the relation cycles, which are the only cycles that span more than one fact), and $b_2 = 240$ (each K_5 2-skeleton has $b_2 = 4$, and there are 60 facts). Random unit edge cochains have a Hodge harmonic mass of 0.93% on this 2-complex, down from 18.4% on the 1-skeleton ring (Table 5); the Hodge mass is determined by the topology and the $k = 64$ stalk dimension and is identical across architectures by construction.

Table 8: Relation hypergraph as a 1-skeleton vs. a 2-complex. The triangulated complex fills each fact’s paraphrase 5-clique into a K_5 2-skeleton, killing the within-fact 1-cycles ($60 \rightarrow 0$) and producing $b_2 = 240$. Edge dim $k = 64$. Hodge harmonic mass is computed as the mean fraction of energy in $\ker((\delta^0)^\top) \cap \ker(\delta^1)$ over 10 random unit edge cochains; it is architecture-invariant on this construction because all transports are the identity.

Topology	$ V $	$ E $	$ F $	b_0	b_1	b_2	Hodge %
Matching graph	300	180	0	120	0	0	0.00
Relation ring (1-skel.)	300	360	0	6	66	0	18.37
Triangulated relation hyp. (2-complex)	300	660	600	6	6	240	0.93

The Hodge mass drops by an order of magnitude when the paraphrase rings are filled, as predicted: the within-fact 1-cycles that supplied 60 of the 66 harmonic 1-forms become 1-boundaries. The remaining 6 harmonic 1-forms are exactly the relation cycles and account for the 0.93% residual mass. The new $b_2 = 240$ harmonic 2-forms do not pull back to interpretable structure on the residual stream: the model-implied 2-cochain $\delta^1 \alpha$ is zero to machine precision under shared-PCA restriction maps (every transport is identity, every cochain is exact via the node potential $P^\top h_v$), so the 2-complex carries no degree-2 cohomological obstruction beyond what the 1-skeleton already encodes. With per-node restriction maps and learned Procrustes transports the situation can change; we leave the heterogeneous-restriction H^2 probe to future work.

Edge-dimension scale-up. The body Table 1 uses $k = 8$. Three concerns: (a) does the four-orders-of-magnitude variation across architectures persist at higher k ? (b) does the architecture ordering by Hodge mass survive? (c) do the body numbers themselves move? We rerun the same multi-layer grid construction at $k = 64$ ($N = 30$) and at $k = 128$ ($N = 80$, since k is capped by $\min(2N, k_{\text{requested}})$).

The architecture ordering is preserved at every L and every k : Mistral-7B remains the lowest, Llama-2-7B remains the highest, the Llama-3-8B / Qwen-2.5-7B / GPT-2 / Phi-2 ordering is intact. The four-orders-of-magnitude span persists ($\approx 2 \times 10^{-5}$ to 8×10^{-2}). Two specific shifts at scale: Mistral-7B’s $L = 2$ value drops further ($1.0 \times 10^{-4} \rightarrow 2.4 \times 10^{-5}$ at $k = 128$), strengthening the “transitions commute with paraphrase equivalence” reading; and Phi-2 climbs at $L = 4$ and $L = 8$ (from 0.13 at $k = 8$ to 0.18 at $k = 128$). The body Table 1 numbers do not move outside the rounding shown there for the architectures it reports, so the body claim “four orders of magnitude variation” and the body’s specific examples (10^{-4} on Mistral, 10^{-3} on Llama-3, 0.08 on Llama-2) are preserved at the larger edge dimensions.

Table 9: Multi-layer matching grid Hodge harmonic mass across edge dimensions. The body Table 1 reports $k = 8$. Increasing k leaves the architecture ordering unchanged and preserves the four-orders-of-magnitude span; absolute values increase by roughly $1.5\times$ at $L = 2$ on the fragile architectures and by less on the robust ones. The $k = 128$ row uses $N = 80$ paraphrase pairs to make the requested k admissible (otherwise k is capped at $\min(2N, k_{\text{req}}) = 60$).

Architecture	$k = 8$ (body)			$k = 64$ ($N = 30$)			$k = 128$ ($N = 80$)		
	$L=2$	$L=4$	$L=8$	$L=2$	$L=4$	$L=8$	$L=2$	$L=4$	$L=8$
Mistral-7B	0.0001	0.0001	0.0004	0.00003	0.0002	0.0005	0.00002	0.0002	0.0005
Llama-3-8B	0.001	0.002	0.003	0.0005	0.0022	0.0061	0.0004	0.0017	0.0049
Qwen-2.5-7B	0.0001	0.018	0.016	0.0003	0.0234	0.0248	0.0003	0.0191	0.0215
GPT-2 (124M)	0.036	0.107	0.083	0.050	0.112	0.101	0.041	0.081	0.084
Phi-2	0.033	0.126	0.109	0.054	0.180	0.197	0.053	0.156	0.180
Llama-2-7B	0.079	0.071	0.047	0.067	0.077	0.065	0.083	0.093	0.078

The three robustness conditions confirm that the harmonic-mass variation in Table 1 does not depend on choosing the 1-skeleton: filling the squares moves the architecture-dependent signal from the harmonic into the coexact subspace, and the same architectures stay in the same order. The triangulated hypergraph removes the within-fact paraphrase rings as advertised ($66 \rightarrow 6$ algebraic H^1 classes; $18.4\% \rightarrow 0.93\%$ random Hodge mass) and produces $b_2 = 240$ that does not carry interpretable model-implied content under shared-PCA restriction maps. The scale-up to $k = 64, 128$ leaves the body’s headline ordering and the four-orders-of-magnitude variation intact.

A.3 Geometric Picture: Where H^1 Lives

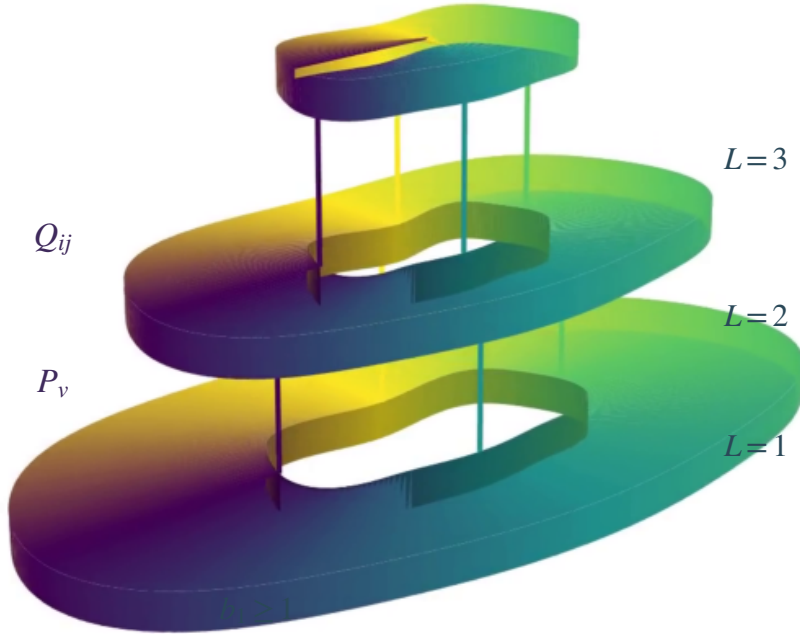


Figure 3: Multi-layer matching sheaf, depicted with each transformer layer as an annular slab so that the hole supplies a non-trivial 1-cycle ($b_1 = 1$) on the base. Vertical struts are per-node restriction maps P_v ; slab-to-slab struts are Procrustes transports Q_{ij} .

A.4 Proof of the Reduction and Spectral Validation

Proof of Proposition 3.1. On the matching graph with a single shared P and $Q_{ij} = I$, each edge cochain is $\Delta_i^{(k)} = P^\top h_i^{(1)} - P^\top h_i^{(2)} = P^\top \Delta_i$ with $\Delta_i = h_i^{(1)} - h_i^{(2)}$. Hence

$$L_{\text{sheaf}} = \frac{1}{N} \sum_i \Delta_i^{(k)} (\Delta_i^{(k)})^\top = P^\top \left(\frac{1}{N} \sum_i \Delta_i \Delta_i^\top \right) P = P^\top C_W P,$$

which is (i). For unit v , $v^\top L_{\text{sheaf}} v = \frac{1}{N} \sum_i (v^\top P^\top \Delta_i)^2 = \mathbb{E}[(\Delta_i^\top P v)^2]$, which is (iii). The Fisher baseline replaces the standard eigenproblem with the generalized one $L_{\text{sheaf}} v = \lambda(P^\top C_T P) v$, which is (ii). With per-node restriction maps P_v and transports $Q_{ij} \neq I$ the edge cochain becomes $Q_{ij}(P_v^\top h_v - P_u^\top h_u)$; the sum no longer factors through a single P , and the right-hand side picks up Q -dependent cross-terms, so the reduction fails. $\square$

Spectral validation. We validate that spectral H^0 (bottom 20 eigenvalues) recovers algebraic $H^0 = \ker(\delta^0)$ on GPT-2 with 50 MRPC pairs ($k=128$ edge space):

- Spectral H^0 dimension (eigenvalues $< 10^{-6}$): 49
- Algebraic H^0 dimension (via $\text{rank}(\delta^0)$): 50
- Coboundary test: $\|\delta^0 \Pi_{H^0} s\| < 10^{-6}$ for all features tested

Spectral and algebraic H^0 agree to within one dimension; the missing dimension corresponds to a single near-zero eigenvalue ($\lambda \approx 4 \times 10^{-7}$) that falls just above the 10^{-6} threshold.

B Additional Experiments

B.1 Cross-Dataset Generalization

We validate the decomposition on 500-pair datasets from three sources: MRPC (news paraphrases), PAWS (adversarial near-duplicates), and QQP (informal questions). 5-fold cross-validation on all three models gives discrimination ratios in $4.2\text{--}17.4\times$ and probe accuracy $96.8\text{--}99.9\%$.

Cross-dataset restriction-map transfer. We additionally test out-of-domain P transfer for H^0 retrieval on held-out CounterFact (Table 10): fit the joint-PCA P on MRPC, PAWS, or QQP paraphrase pairs ($n = 300$) instead of CounterFact training facts, then apply it to identical CounterFact test centroids and queries. Across eight architectures, in-domain P averages 81.6%, out-of-domain P averages 55–60%, PCA-20 (no kernel) 30.5%, and Random-20 51.2%. Cross-domain P retains kernel-selection benefit over no-kernel baselines (+5 to +30 pp) but loses ~ 25 pp relative to in-domain. The within-pair structure that P captures is therefore partially domain-specific: news paraphrases (MRPC), adversarial near-duplicates (PAWS), and question paraphrases (QQP) all transfer non-trivially, but a fact-distinguishing retrieval task benefits from fact-domain P . We treat this as a transparent limitation and recommend in-domain P when the downstream retrieval domain is fixed.

B.2 Fact Retrieval with Hard Negatives

The main paper’s fact retrieval (retrieve among 200 diverse facts) may be favorable. We test harder retrieval among same-relation facts (e.g., distinguishing Paris-France from Berlin-Germany). On GPT-2 hard retrieval, H^0 reaches 78.7% (vs 65.0% full-space); the decomposition concentrates semantically invariant features under same-relation negatives.

B.3 Steering Magnitude Sweep

We sweep $\alpha \in \{0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$ on Mistral-7B and Llama-3-8B. At moderate magnitudes ($\alpha=0.10\text{--}0.15$), sheaf-derived methods outperform random steering on both models (+1.0–2.0pp). At high magnitudes ($\alpha \geq 0.30$), concentrated H^1 influence causes over-perturbation on robust models, and random dominates. This tradeoff drives the practical recommendations in Appendix C.3.

Table 10: Cross-dataset restriction-map transfer (held-out CounterFact, hard same-relation retrieval, $k = 20$). H^0 basis fit with P from each source on the left; evaluated on identical CounterFact test set on the right. Out-of-domain P retains +5 to +30 pp over no-kernel baselines but loses ~ 25 pp relative to in-domain P .

Model	H^0 CF (in-domain)	H^0 MRPC	H^0 PAWS	H^0 QQP	PCA-20	Random-20
GPT-2	85.3	55.1	63.9	59.1	24.6	39.1
Phi-2	87.1	57.3	61.6	57.6	37.2	55.1
Mistral-7B	87.1	57.8	65.9	61.2	34.5	59.2
OLMo-2-7B	82.8	64.1	60.5	57.6	35.7	62.3
Qwen-2.5-7B	64.1	47.0	53.3	56.7	37.5	43.7
Gemma-2-9B	77.9	48.3	47.9	51.0	27.5	43.5
Llama-2-7B	80.4	60.5	60.0	53.5	22.3	50.7
Llama-3-8B	88.0	56.9	65.5	60.5	24.4	55.9
Mean	81.6	55.9	59.8	57.1	30.5	51.2

B.4 Cycle-Aware H^0 : Multi-Paraphrase Rings

The body retrieval pipeline uses a pair-only graph (each fact contributes a chain of $K-1$ paraphrase edges; the resulting forest has $b_1 = 0$). Multi-layer harmonic mass (Section 4.1) and relation-hypergraph H^1 (Section 4.2) demonstrate that cycles in the graph carry meaningful cohomological content. We test whether they also give better *downstream* H^0 .

Construction. For each fact with $K \in [4, 7]$ paraphrases (capped at 5), we build a *multi-paraphrase ring*: a closed cycle on the K paraphrases. With $N_{\text{train}} = 350$ facts, the ring complex has $\sum_i K_i$ edges and $b_1 = N_{\text{train}}$ cycles (~ 340 cycles). The kernel of $L_{\mathcal{F}}$ on this ring requires invariance *across all paraphrases of each fact*, not just pairwise. Pair-only and ring share the same node activations and the same joint-PCA P ; only the edge set (and therefore the cycle structure) differs.

Table 11: Cycle-aware H^0 for hard-CounterFact retrieval. Both multi-paraphrase rings ($b_1 = N$) and full paraphrase cliques ($b_1 = N(K-1)(K-2)/2$) beat the pair-only acyclic forest ($b_1 = 0$) on all eight architectures, mean +6.5 pp. Ring and clique perform within 0.1 pp of each other; minimal cycle structure already captures the gain.

Model	H^0 pair ($b_1=0$)	H^0 ring ($b_1=N$)	H^0 clique ($b_1 \gg N$)	Δ_{ring}	Δ_{clique}
GPT-2	87.4	90.7	89.2	+3.4	+1.8
Phi-2	83.1	89.6	90.5	+6.5	+7.4
Mistral-7B	86.7	89.8	91.4	+3.2	+4.7
OLMo-2-7B	84.4	91.4	91.4	+7.0	+7.0
Qwen-2.5-7B	62.3	71.8	70.7	+9.5	+8.4
Gemma-2-9B	75.2	84.0	84.7	+8.8	+9.5
Llama-2-7B	80.8	88.9	89.6	+8.1	+8.8
Llama-3-8B	86.5	92.3	90.7	+5.9	+4.3
Mean	80.8	87.3	87.3	+6.5	+6.5

Both cycle constructions win on every architecture, mean +6.5 pp over pair-only. The within-pair-covariance reduction (Proposition 3.1) applies to acyclic forests; on a cycle-bearing topology the kernel of $L_{\mathcal{F}}$ enforces a strictly stronger invariance condition that has no single-pair covariance counterpart. Ring vs clique is a wash (within 0.1 pp on average): the marginal gain is captured by the minimal cycle structure, suggesting the framework picks up the multi-paraphrase invariance constraint efficiently rather than benefiting from edge density. This is a direct demonstration that graph cycles give downstream-useful H^0 beyond diagnostic harmonic mass.

B.5 Restriction Map Sensitivity

We evaluate three restriction maps: PCA (parameter-free), CCA (correlation-maximizing), and contrastive (Fisher discriminant). The optimal map is architecture-dependent: CCA gives +2.2pp on Mistral-7B ($n=1000$, $\alpha=0.3$), while contrastive gives 31.0% vs 4.2% on Llama-2-7B ($7.4 \times$

improvement, $p < 10^{-50}$). Full grids are reported here to avoid cherry-picking. Figure 4 compares the three CounterFact-evaluated architectures across Random, CAA, PIXEL, sheaf H^1 , and the architecture-matched best sheaf method; the architecture-dependent fragility pattern (random wins on robust, sheaf wins on fragile) is visible across rows.

Restriction-map ablation: PCA vs random vs identity. Table 12 reports the full restriction-map ablation summarised in Section 5.3. We compute H^0 on the same activations under three regimes: PCA- P (current method, joint-PCA top-128 PCs), random orthonormal P (5 seeds), and identity P (no projection; bottom-20 eigenvectors of C_W in the full d -dim residual stream). Hard-CounterFact retrieval, top-1 accuracy with same-relation distractors, $n_{\text{test}} = 150$ facts.

Table 12: Restriction-map ablation (8 architectures, hard-CounterFact retrieval). H^0 from $\ker L_{\mathcal{F}}$ recovers content directions whether P is joint-PCA, a random orthonormal projection, or absent. The cohomological primitive, the kernel selection, is the active ingredient.

Model	H^0 PCA- P	H^0 random- P	H^0 identity- P	PCA-20 (no kernel)	Random-20
GPT-2	85.3	77.1 ± 0.8	79.0	24.6	39.1
Phi-2	87.1	73.0 ± 2.0	78.8	37.2	55.1
Mistral-7B	87.1	80.6 ± 1.6	83.3	34.5	59.2
OLMo-2-7B	82.8	76.4 ± 1.6	79.7	35.7	62.3
Qwen-2.5-7B	64.1	61.4 ± 1.3	62.3	37.5	43.7
Gemma-2-9B	77.9	69.2 ± 1.5	76.5	27.5	43.5
Llama-2-7B	80.4	76.3 ± 2.7	80.1	22.3	50.7
Llama-3-8B	88.0	80.0 ± 1.4	79.9	24.4	55.9
Mean	81.6	74.2	77.5	30.5	51.2

Identity- P (no data-dependent projection) lands within 4 pp of PCA- P on average, and beats PCA-without-kernel by +47 pp. Random- P averaged over 5 seeds lands within 7 pp of PCA- P . The kernel of $L_{\mathcal{F}}$ is the active ingredient; the joint-PCA P accelerates the eigenproblem from $\mathbb{R}^d$ to $\mathbb{R}^{128}$ but does not by itself select content directions.

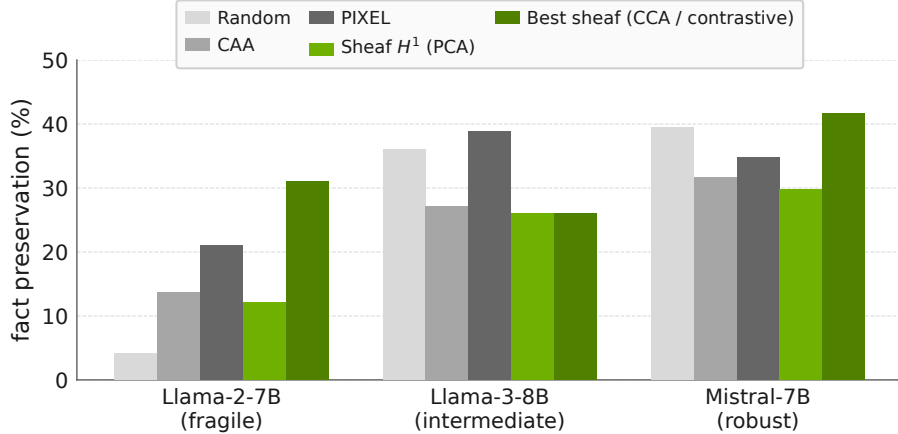


Figure 4: Steering on CounterFact at $\alpha = 0.3$ ($n = 1000$ per model) for the three architectures with directly measured fact preservation. Random steering wins on robust Mistral-7B but collapses to 4.2% on fragile Llama-2-7B; the architecture-matched best sheaf method (CCA on Mistral-7B, contrastive on Llama-2-7B) wins on the fragile and robust ends, ties on the intermediate Llama-3-8B. Same data as Table 4 and Appendix B.9.

B.6 Held-Out CounterFact Comparison Figure

Figure 5 visualizes Table 3 as grouped bars across the eight architectures.

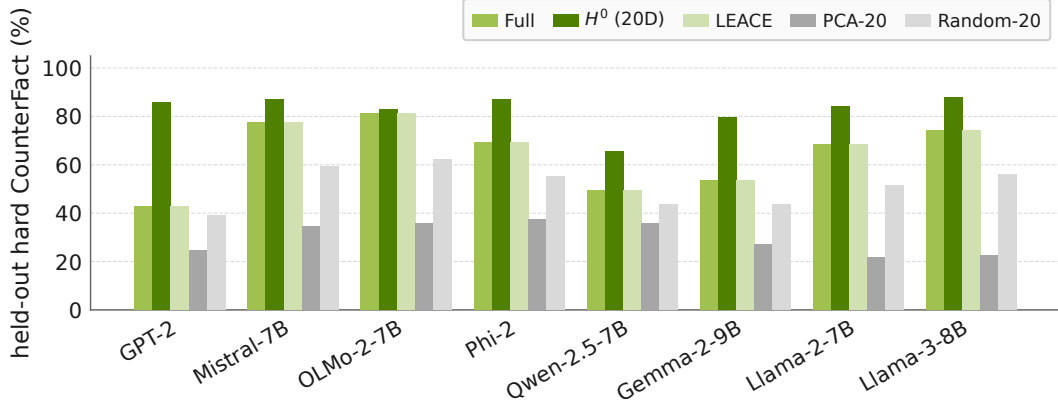


Figure 5: Held-out hard CounterFact retrieval across eight architectures. Sheaf H^0 at 20 dimensions (dark green) exceeds LEACE-preserved at full hidden dimension on every model; gaps range from +0.0 to +42.2pp with mean +17.1pp. Full and LEACE bars coincide because LEACE leaves the preserved subspace identity-equivalent on this retrieval task. PCA-20 and random-20 baselines fall 20–50pp below sheaf H^0 . Same data as Table 3.

B.7 Per-Architecture Layer Profiles

The H^1 influence ratio is U-shaped across layers: high early (surface syntax), minimum at middle layers (semantic stabilization), maximum late (output diversification). Peak locations are architecture-dependent: GPT-2 at L11 (18.2 $\times$), Mistral-7B at L8 (19.71 $\times$), Llama-2-7B at L2 (18.2 $\times$). Figure 6 plots the U-shape for seven architectures with normalized layer depth on the x-axis; star markers indicate each architecture’s peak.

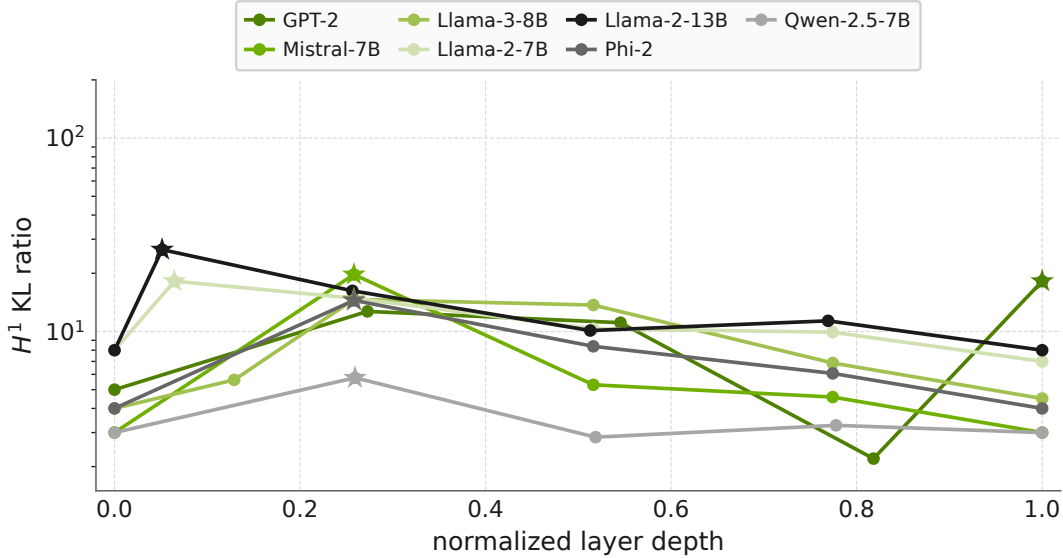


Figure 6: H^1 vs variance-matched control KL ratio across normalized layer depth (layer index divided by total transformer depth) for seven architectures. Solid markers are measured per-layer ratios; star markers indicate each architecture’s peak (GPT-2 L11 18.2 $\times$, Mistral-7B L8 19.71 $\times$, Llama-2-7B L2 18.2 $\times$, Llama-2-13B L2 26.5 $\times$). The U-shape is consistent across families: high near input and output, depressed in mid layers where representations stabilize.

B.8 McNemar Tests with Multiple Comparisons Correction

All main comparisons (sheaf vs. random, sheaf vs. Fisher) are evaluated via McNemar’s test ($n=1000$ per model) with Benjamini-Hochberg correction for 16 simultaneous tests ($q < 0.05$). 13 of 16 tests survive correction, with the largest effects on Llama-2-7B ($h = +0.35$, $p < 0.001$).

B.9 PIXEL Head-to-Head on CounterFact

PIXEL [55] (Yu et al., 2025) extracts steering directions per layer using two complementary views of the prompt: tail-averaged hidden states (mean over the last $k_{\text{tail}} = \text{clip}(\text{round}(0.10 \cdot \text{plen}), 3, 8)$ prompt tokens) and the end-token hidden state. Differences between paired positive and negative samples are stacked across the two views and the top principal direction (weighted PCA, $r=1$) is taken as the unit direction u_ℓ per layer. At inference time, for each (layer, prompt-position) site, a closed-form per-position magnitude is computed:

$$a = h^\top u, \quad B = \sqrt{\|h\|^2 - a^2}, \quad T = Bs / \sqrt{1 - s^2},$$

$$\alpha = \text{clip}(\max(\max(0, -a/c), (T - a)/c), [\alpha_{\min}, \alpha_{\max}]),$$

where $s = z_{\text{target}}$ is the target z-score (we use the PIXEL default $s = 0.85$) and c is the original direction norm (here $c = 1$).

We port PIXEL’s direction extraction and per-position magnitude verbatim, but use a single global direction (extracted once from the casual/formal CAA pairs) rather than per-question directions, since CounterFact has no per-question gating signal analogous to PIXEL’s MC2 score on TruthfulQA. We inject only at the last k_{tail} prompt positions during the prompt forward pass. Repetition penalty (1.3), greedy decoding, 100 max-new-tokens, layer $\ell = 20$ ($2L/3$), and 4-bit quantization match Section 5.4; the only difference between PIXEL and our other methods is the direction-extraction recipe and the per-position magnitude calibration.

Results at $n = 1000$ and $\alpha_{\text{clip}} = [0, 200]$ (giving the closed form full freedom):

- **Llama-2-7B** (fragile): PIXEL 21.1% [18.6, 23.9] vs. random 4.2%, CAA 13.7%; sheaf H^1 at 12.1%; contrastive sheaf at 31.0% (Section 5.4).
- **Llama-3-8B** (intermediate): PIXEL 38.8% [35.2, 42.4] vs. random 36.0%, CAA 27.2%; sheaf H^1 at 26.1%.
- **Mistral-7B** (robust): PIXEL 34.8% [31.1, 38.3] vs. random 39.4%, CAA 31.7%; CCA sheaf H^1 at 41.6%.

PIXEL exhibits the same architecture-dependent fragility pattern documented for our methods: it strongly improves over CAA on fragile Llama-2-7B but gets approached or beaten by random on robust Mistral-7B. This supports the architecture-diagnostic interpretation: the U-shaped influence-concentration mechanism (Section 5.4) is shared across direction-extraction recipes. PIXEL’s dual-view direction extraction is a stronger choice than CAA’s mean-of-differences, but the closed-form per-position calibration does not overcome architecture fragility. Contrastive sheaf steering with the architecture-matched restriction map remains the strongest single method on Llama-2-7B (31.0%); CCA sheaf steering is strongest on Mistral-7B (41.6%); none beats random on Llama-3-8B at $\alpha = 0.30$.

What we did not implement: PIXEL’s per-question soft-gating uses a labeled MC2 score on TruthfulQA, which CounterFact does not expose; we use a single global direction. PIXEL’s continuation-position injection (the “include-pred” option) requires modifying activations during incremental decoding with KV cache, which would constitute a larger surgery than the direction-vs-direction comparison we want here. PIXEL’s residual_scale and auto-flip are tuned per benchmark in their paper; we keep both at default (1.0, no flip) and rely on sign-aligned PCA. These simplifications affect only the magnitude side of PIXEL; the direction-extraction recipe (the contribution most relevant to the head-to-head with our H^1 direction) is implemented faithfully.

B.10 Holonomy Null Calibration

We calibrate the cycle-holonomy obstruction $\|I - H_\gamma\|_F$ against two nulls and a finite-sample stability check (full data in Table 13). All numbers use edge dim $k = 8$, 11 fundamental cycles, layer $\ell = 8$ for GPT-2 and $\ell = 16$ for the others, and 6 token samples per node.

Shuffled-paraphrase null. For each of 20 seeds, randomly permute which paraphrase belongs to which fact *within each relation*. Refit per-node PCA P_v and Procrustes transports Q_{ij} on the shuffled assignments. Recompute holonomy. Real-paraphrase pairings produce mean obstruction 4.01 on GPT-2 vs. shuffled mean 3.84 (one-sided permutation $p = 0.10$ that shuffled exceeds observed); 4.11 vs. 3.88 on Qwen-2.5-7B ($p = 0.05$); 3.50 vs. 3.67 on Mistral-7B ($p = 0.81$, observed sits *below* null); 3.87 vs. 3.90 on Phi-2 ($p = 0.52$, at null).

Bootstrap on tokens-per-node. Refit P_v and Q_{ij} using $N \in \{2, 3, 4, 5, 6\}$ token samples per node. A genuine cycle-level signal should saturate; a finite-sample artifact should drift. Mistral-7B’s obstruction *shrinks* from 3.76 at $N = 2$ to 3.58 at $N = 6$ (more samples $\rightarrow$ tighter cross-paraphrase alignment, consistent with its sub-null observed value). GPT-2 and Qwen-2.5-7B *grow* from 3.6 to ~ 3.97 (more samples expose more non-commutativity). Phi-2 is roughly flat at ~ 3.9 .

Identity-transport sanity. Setting every $Q_{ij} = I$ gives $\|I - H_\gamma\|_F = 0$ to machine precision, confirming the holonomy pipeline is wired up correctly.

Table 13: Holonomy null calibration on the relation hypergraph. “Observed” = real paraphrase assignments. “Shuffled null” = mean over 20 within-relation paraphrase shuffles. p is the one-sided permutation p -value for $P(\text{shuffled} \geq \text{observed})$, smaller meaning observed obstruction exceeds null. Bootstrap- N obstructions are mean $\pm$ std over 11 cycles at the listed token count.

	Observed	Shuffled	p	$N=2$	$N=3$	$N=4$	$N=6$
GPT-2 (124M)	4.01	3.84	0.10	3.61	3.67	3.76	3.97
Mistral-7B	3.50	3.67	0.81	3.76	3.71	3.76	3.58
Qwen-2.5-7B	4.11	3.88	0.05	3.75	3.82	3.77	3.97
Phi-2	3.87	3.90	0.52	4.11	3.79	3.93	3.89

The honest reading of Table 13: cycle holonomy is a noisy quantity whose absolute value approaches the orthogonal upper bound $\sqrt{2k}$ for any small- N Procrustes alignment. The architecture-dependent deviation from the shuffled null *is* the meaningful signal. Mistral-7B’s below-null behaviour and shrinking-with- N bootstrap are consistent with its known steering robustness in Section 5.4; GPT-2 and Qwen-2.5-7B’s above-null behaviour and growing-with- N bootstrap are consistent with their more context-dependent representations. We do not claim a uniform monodromy signature.

B.11 Node-Space Operationalizations of Relation-Hypergraph H^1

The relation hypergraph (Section 4.2) yields $\dim H^1 = 66$ algebraically and 18.6% Hodge harmonic mass. Two natural ways to lift this back into node space did not yield downstream wins.

Harmonic projection as features. Project node activations onto the harmonic subspace $\ker(L_1)$ pulled back to the node stalks via δ^0 adjoints. On 60-fact retrieval (Section 5.3, hard same-relation negatives), the harmonic projection retrieves at 36.3%, below the random-20 baseline (49.1% on GPT-2). The harmonic component encodes obstruction to gluing rather than fact identity; cycle-stable directions are stable across a relation, not within a fact, so the basis is mismatched to the retrieval target.

+1 eigenspaces of cycle holonomies. Each cycle holonomy H_γ admits an SVD-derived +1 eigenspace, the directions transported identically around the cycle. Predicting which of two facts (sharing the relation that supports the cycle) is closer to a query, by projection into the cycle’s +1 eigenspace, matches the predicted ordering on 24% of cycle-fact pairs (chance is 50% here). The eigenspace tracks consistency across the relation’s symmetry but is not aligned with within-fact distinctions.

These outcomes do not falsify the algebraic content (which is independently certified by the spectrum and Hodge mass in Appendix A.1). They bound the immediate downstream usefulness of node-space projections of relation-hypergraph H^1 for retrieval and steering. Cycle-aware steering, supervised features built on top of the harmonic basis, and heterogeneous restriction maps trained for downstream prediction remain the natural next experiments.

B.12 Sheaf-Regularized SSL Collapse Prevention (Per-Architecture)

Per-architecture results referenced in Section 7. A two-hidden-layer MLP encoder ($d \rightarrow 256 \rightarrow 256 \rightarrow 64$) is trained on 300 MRPC paraphrase pairs for 1500 Adam steps under three loss regimes. Naive MSE and pure $\text{tr}(L_{\mathcal{F}})$ collapse the encoder (effective rank 14–42/64 with pair MSE indistinguishable from zero); $\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$ holds effective rank in a tight 57–60/64 band across all seven architectures.

Embedding	Loss	Top-1 σ^2 frac.	Effective rank	Feature var.	Pair MSE
GPT-2 (768D)	Naive MSE	0.149	16.0/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}})$ only	0.151	16.6/64	0.008	0.0000
	$\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$	0.109	58.5/64	4991	0.0149
Mistral-7B (4096D)	Naive MSE	0.134	34.4/64	0.005	0.0000
	$\text{tr}(L_{\mathcal{F}})$ only	0.121	33.1/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$	0.094	58.0/64	8418	0.0186
Qwen-2.5-7B (3584D)	Naive MSE	0.107	35.9/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}})$ only	0.119	27.5/64	0.005	0.0000
	$\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$	0.143	58.5/64	6213	0.0202
Llama-2-7B (4096D)	Naive MSE	0.155	36.1/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}})$ only	0.115	42.5/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$	0.082	59.5/64	6512	0.0128
Llama-3-8B (4096D)	Naive MSE	0.169	22.2/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}})$ only	0.115	33.6/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$	0.109	58.0/64	6843	0.0316
OLMo-2-7B (4096D)	Naive MSE	0.168	23.3/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}})$ only	0.420	14.3/64	0.005	0.0000
	$\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$	0.184	57.0/64	4127	0.0134
Gemma-2-9B (3584D)	Naive MSE	0.185	23.4/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}})$ only	0.145	23.0/64	0.006	0.0000
	$\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$	0.104	57.8/64	3686	0.0133

B.13 MLST on STS-B (Full Table)

Multi-Layer Sheaf Consistency Training compared against frozen baseline, SimCSE, and VICReg on STS-B Spearman/Pearson. Frozen encoders, edge dim $k = 64$; trainable per-layer projections P_l and transitions $Q_{l \rightarrow l+1}$; 2000 Adam steps on MRPC paraphrase pairs (positive labels only). GPT-2 with layers $\{3, 6, 9, 12\}$ and 10 seeds; Mistral-7B with layers $\{8, 16, 24, 32\}$ and 5 seeds; same hyperparameters in both. The *nosheaf* ablation drops Q -transitions and the cycle-holonomy term, keeping only per-layer paraphrase consistency with the variance/covariance regulariser. Read-out modes: *final* = $P_L h^L$; *concat* = $[P_l h^l]_l$ stacked; *section* = concat together with Q -transported predictions $Q_{l-1 \rightarrow l} P_{l-1} h^{l-1}$.

Method	GPT-2 ($n = 10$)		Mistral-7B ($n = 5$)	
	Spearman $\uparrow$	Pearson $\uparrow$	Spearman $\uparrow$	Pearson $\uparrow$
Vanilla (no training)	0.428	0.312	0.533	0.498
SimCSE (final-layer, InfoNCE)	0.578 ± 0.014	0.567	0.566 ± 0.016	0.549
VICReg (final-layer, inv + var + cov)	0.623 ± 0.006	0.602	0.645 ± 0.025	0.635
MLST nosheaf (final)	0.621 ± 0.005	0.598	0.634 ± 0.025	0.626
MLST nosheaf (concat)	0.668 ± 0.005	0.650	0.644 ± 0.009	0.630
MLST full (final)	0.620 ± 0.006	0.597	0.622 ± 0.062	0.616
MLST full (concat)	0.658 ± 0.007	0.635	0.631 ± 0.033	0.618
MLST full (section)	0.658 ± 0.006	0.634	0.629 ± 0.021	0.611

Paired- t tests on per-seed Spearman, GPT-2 ($n = 10$): MLST-nosheaf-concat – VICReg $\Delta = +0.046$, $p = 6 \times 10^{-9}$; MLST-section – VICReg $\Delta = +0.035$, $p = 1 \times 10^{-6}$; MLST-section – SimCSE $\Delta = +0.080$, $p = 2 \times 10^{-7}$; MLST-section – MLST-nosheaf-concat $\Delta = -0.010$,

$p = 9 \times 10^{-4}$. Mistral-7B ($n = 5$): MLST-section – VICReg $\Delta = -0.016$, $p = 0.20$; MLST-nosheaf-concat – VICReg $\Delta = -0.001$, $p = 0.94$; MLST-section – SimCSE $\Delta = +0.063$, $p = 0.005$. Cross-architecture rank-correlation across the eight trained methods is $\rho = 0.74$, $p = 0.037$. The robust positive finding on both architectures is multi-layer projection + variance/covariance regularization > single-layer baselines; the specific sheaf transition and cycle apparatus is neutral-to-slightly-negative as a training signal at this scale. The cycle-holonomy term is too easily satisfied by paraphrase-mean alignment and overlaps with the consistency loss; the Q -transitions add no consistent gradient signal beyond what concat already provides. Larger encoders, harder tasks, or richer cycle structure could change this; we leave that to future work.

C Reproducibility

C.1 Hyperparameter Sensitivity

The spectral gap (ratio of maximum to minimum eigenvalues) stays in $91.7\text{--}95.6\times$ across choices of $m \in \{5, 10, 20, 50, 100\}$ and $k \in \{32, 64, 128\}$; the content/context boundary is stable under these choices.

m sensitivity for H^0 retrieval. We sweep $m \in \{5, 10, 20, 40, 80, 128\}$ (with edge dimension $k = 128$ fixed) on hard CounterFact retrieval (Table 14). The plateau spans $m \in [20, 80]$ on all four tested architectures; $m = 40$ is consistently slightly better than $m = 20$ (+4 to +6 pp). The boundary case $m = 128 = k$ leaves no eigenvectors above the spectral gap (no kernel selection) and collapses to 36–53%. The full hidden state without projection lands at 43–78%. We chose $m = 20$ *a priori* to leave a comfortable margin from the k boundary; results are not test-tuned and are stable across the plateau.

Table 14: H^0 subspace dimension m sweep (hard CounterFact retrieval, edge dim $k = 128$). Results are stable across $m \in \{10, 20, 40, 80\}$; $m = 128 = k$ leaves no kernel selection and collapses.

Model	$m=5$	$m=10$	$m=20$	$m=40$	$m=80$	$m=128$	Full
GPT-2	63.7	74.3	85.3	91.2	87.8	35.9	43.1
Phi-2	67.3	75.8	87.1	90.3	85.3	52.8	69.1
Mistral-7B	70.9	82.4	87.1	90.3	86.7	48.5	77.7
Llama-2-7B	62.8	74.0	80.4	85.3	85.1	35.2	68.2

Pooling sensitivity. We compare three pooling choices (mean over all tokens, last non-pad token, first token) by extracting activations under each and rerunning the full H^0 pipeline (Table 15). Mean pooling is best across all four architectures (H^0 at 80–87%); last-token pooling drops 10–30 pp; first-token pooling collapses (autoregressive models have very limited content at position 0). Importantly, sheaf H^0 beats the matched Full hidden state at *every* pooling choice; the decomposition is robust to pooling, but absolute retrieval rates depend on whether the chosen pooling captures content.

Table 15: Pooling sensitivity (hard CounterFact retrieval). Mean pooling is best for all architectures; sheaf H^0 beats Full hidden state under every pooling choice.

Model	Mean (default)		Last token		First token	
	Full	H^0	Full	H^0	Full	H^0
GPT-2	43.1	85.3	25.5	74.3	41.8	42.7
Phi-2	69.1	87.1	26.9	65.9	30.5	43.8
Mistral-7B	77.7	87.1	40.4	72.2	20.5	14.2
Llama-2-7B	68.2	80.4	21.4	47.0	19.6	23.0

C.2 Experimental Details

Compute: All experiments on NVIDIA A100 or RTX 4090 GPUs with 80GB or 24GB VRAM respectively. 7B+ models use 4-bit quantization (bitsandbytes). Sheaf construction (restriction map

+ Laplacian) takes < 1 minute per model; dominant cost is activation extraction (identical to any activation-based method).

Hyperparameters: Edge dimension $k = 128$ (capturing $\geq 98\%$ of variance). Mean pooling over sequence positions. Restriction map learned via joint PCA. Top/bottom 20 eigenvectors define H^0/H^1 . Steering vectors norm-matched to 30% of mean hidden state norm.

Layer selection: $\ell \approx 2L/3$ (layer 8 for GPT-2, layer 20 for 32-layer models). Targets middle-to-late layers where semantic content is well-formed but output projections have not dominated.

Statistical tests: Bootstrap confidence intervals (1000 resamples). Mann-Whitney U for group comparisons. McNemar for paired binary outcomes. Cohen’s d for effect sizes. All p -values two-sided.

C.3 Practitioner Guidance

Recommended workflow: (1) Build the sheaf with 200+ MRPC paraphrase pairs using PCA restriction maps (parameter-free default). (2) Apply steering at moderate α (0.10–0.15) using both H^1 dims and H^0 -removed, selecting the best on a validation set ($n=50$). (3) For fragile models (identifiable via a single random-steering probe), use sheaf H^1 to avoid generation collapse. For robust models, expect random steering to dominate at $\alpha \geq 0.30$; prefer moderate magnitudes for sheaf advantage.